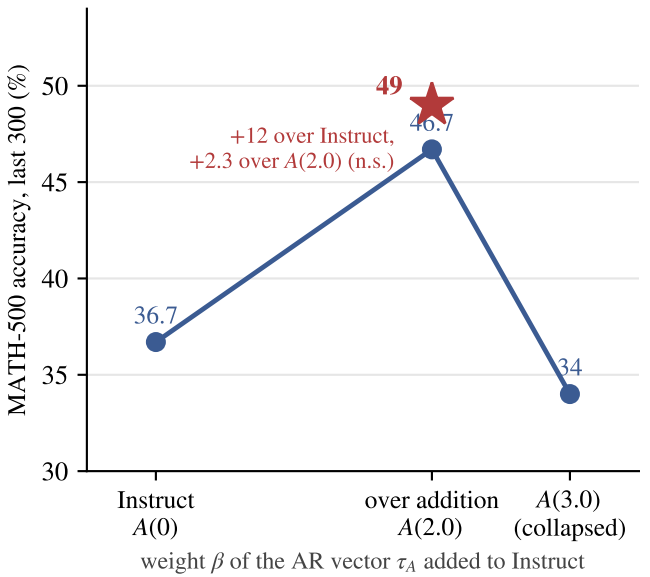
Over distillation on the addition line: the over-added model finishes but is not more precise — keep the student's tokens where it is confident and take the teacher's only where it is unsure

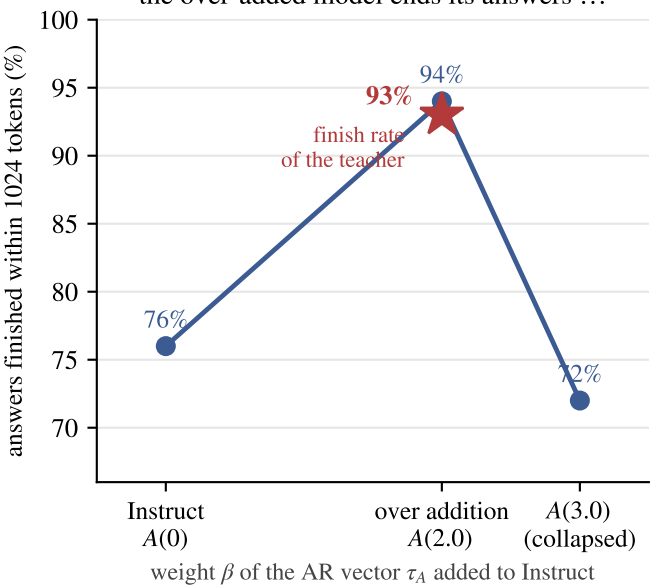
one model on the addition line (1× decode)

gate Ins ⇐$A(2.0)$ @ θ=0.8: Instruct decodes, $A(2.0)$ supplies the tokens where Instruct is unsure (2× decode)

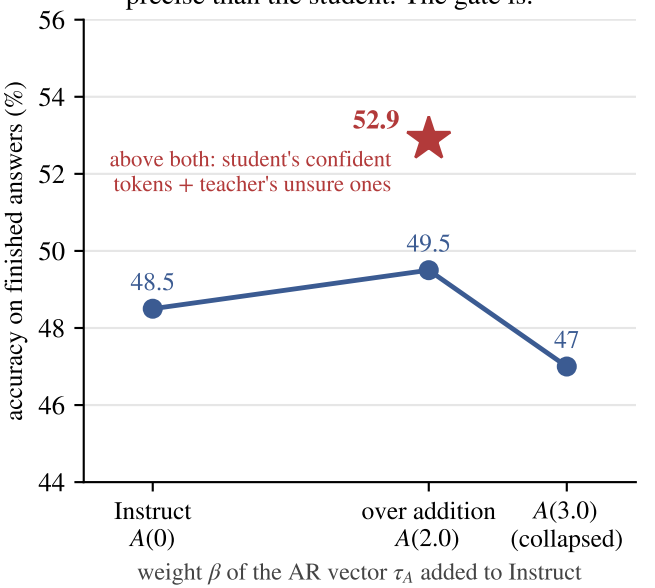
(a) Adding the AR vector: accuracy rises to $A(2.0)$ and collapses at $A(3.0)$ …



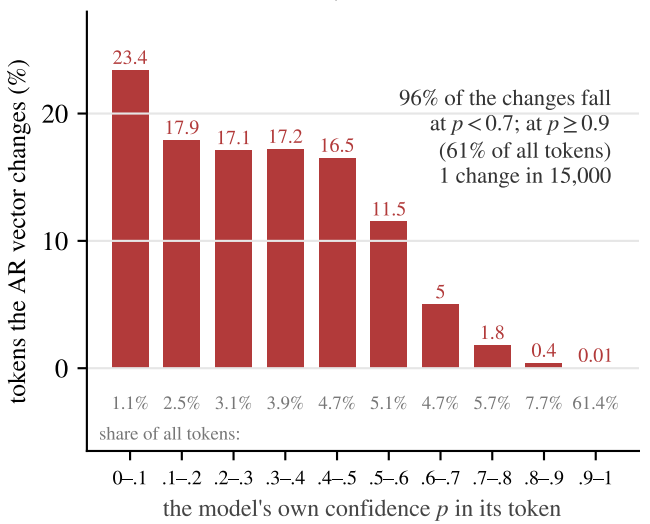
(b) … the whole gain is finish rate: the over-added model ends its answers …



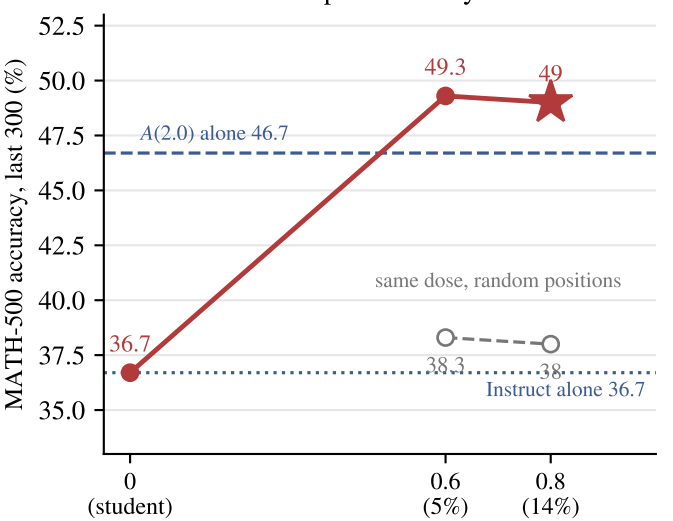
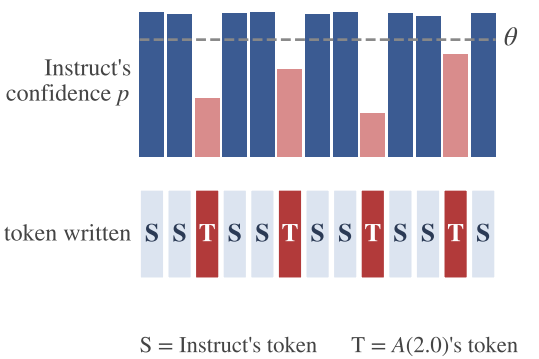
(c) … but per finished answer it is no more precise than the student. The gate is.



(d) Why ask where the student is unsure: the AR vector rewrites a token almost only when the model is unsure about it (A2D, teacher-forced, GSM8K)



(e) Threshold: the teacher's level is reached by $\theta = 0.6$–0.8 with 5–14% of the tokens; the same dose at random positions stays at the student



(f) The gate: Instruct picks every position left to right and writes its own token where $p \geq \theta$; where $p < \theta$ the teacher's token is written